

Cornell Bowers C-IS

College of Computing and Information Science

Deep Learning

Week 7: Diffusion Models II

Thanks to

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Adhitya Polavaram
Sofian Zalouk

Logistics

- **HW4** released - due Sunday (03/23)
 - still have two slip days - late due date (03/25)
 - Shorter assignment on VAEs and Diffusion
- **P4** released soon - due **after** Spring Break.
- **Practice Midterm** was released early this week!
- Midterm next Thursday (03/27)
 - Makeup Exams details to be announced this week.
 - Location and time will be posted on Ed

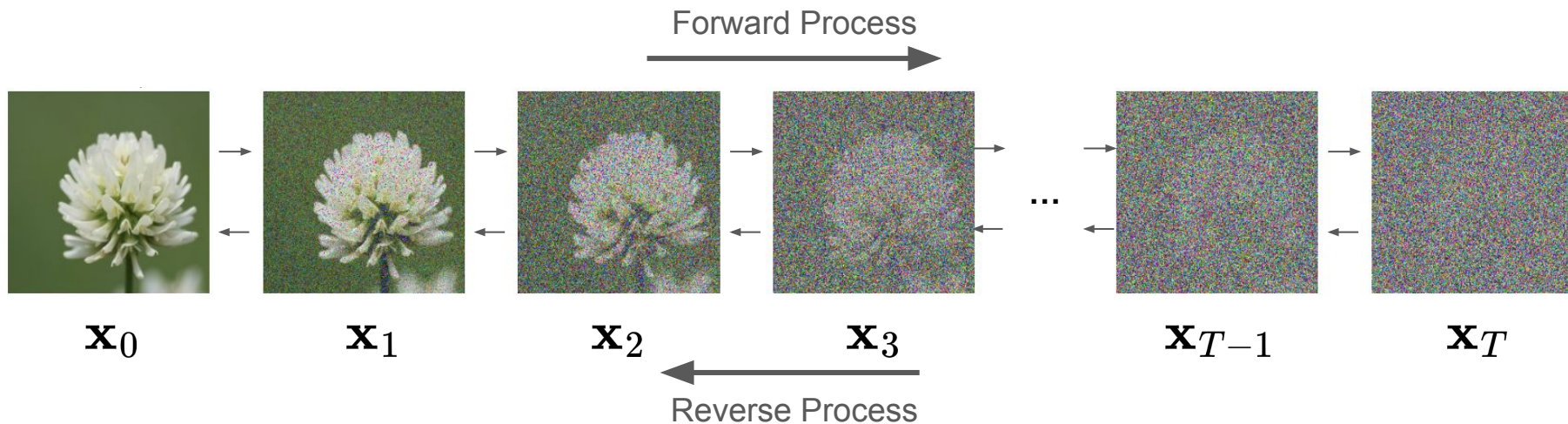
Today's lecture

- Brief review of Diffusion Models
- **Score-based** generative modeling offers a different perspective on DDPM
- **Conditional diffusion** models can be used to generate samples conditioned on other text or image
- **Stable diffusion** is performed in the latent spaces of pre-trained autoencoders

Denoising Diffusion Models

Denoising diffusion models consist of two processes:

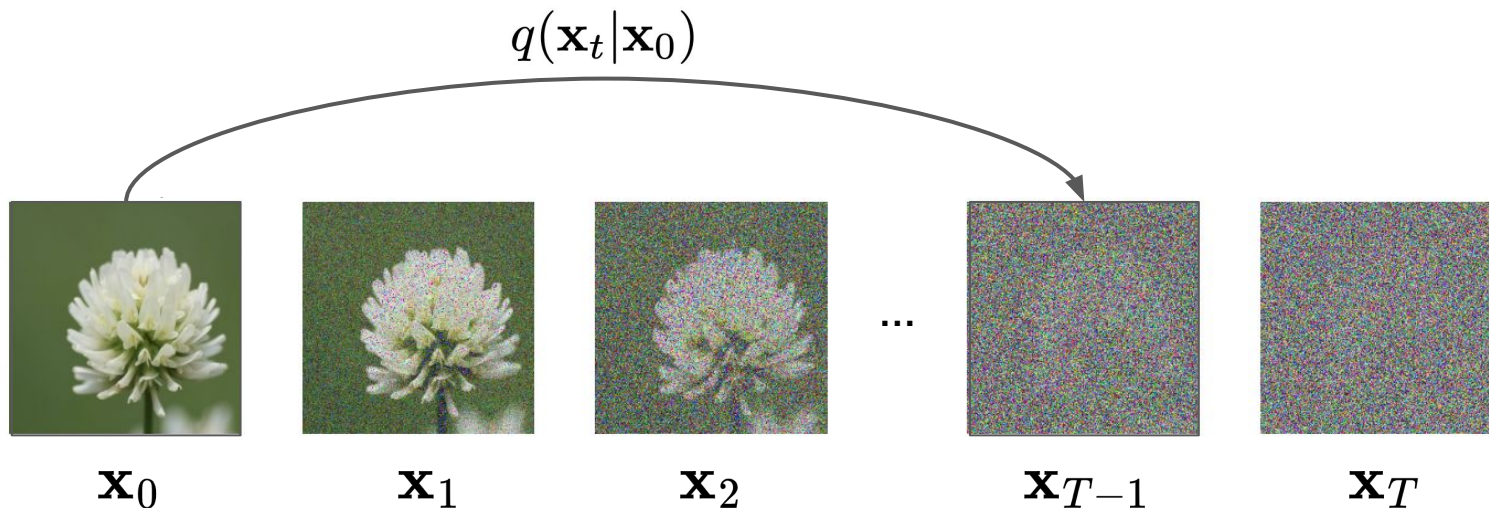
- Forward diffusion process that gradually adds noise to input
- Reverse denoising process that learns to generate data by denoising



Details: Forward Process

Can sample $\mathbf{x}_t$ in closed-form as $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \bar{\alpha}_t \in (0, 1)$$



Aside: Noise Schedules

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \bar{\alpha}_t \in (0, 1)$$

- Define the noise schedule in terms of $\bar{\alpha}_t \in (0, 1)$
 - Some monotonically decreasing function from 1 to 0
- Cosine Noise schedule:

$$\bar{\alpha}_t = \cos(.5\pi t/T)^2$$

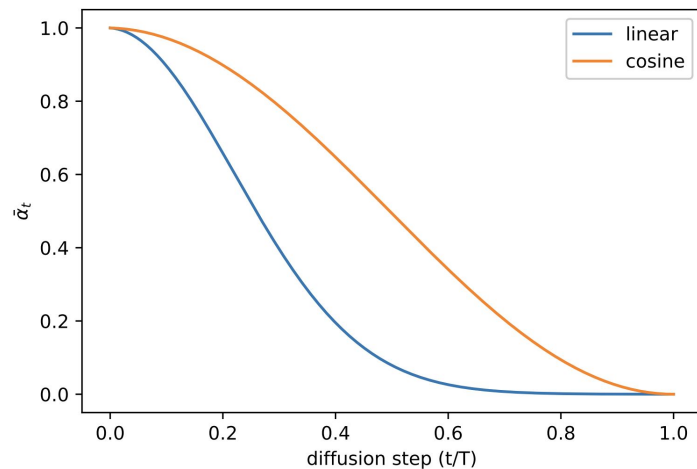
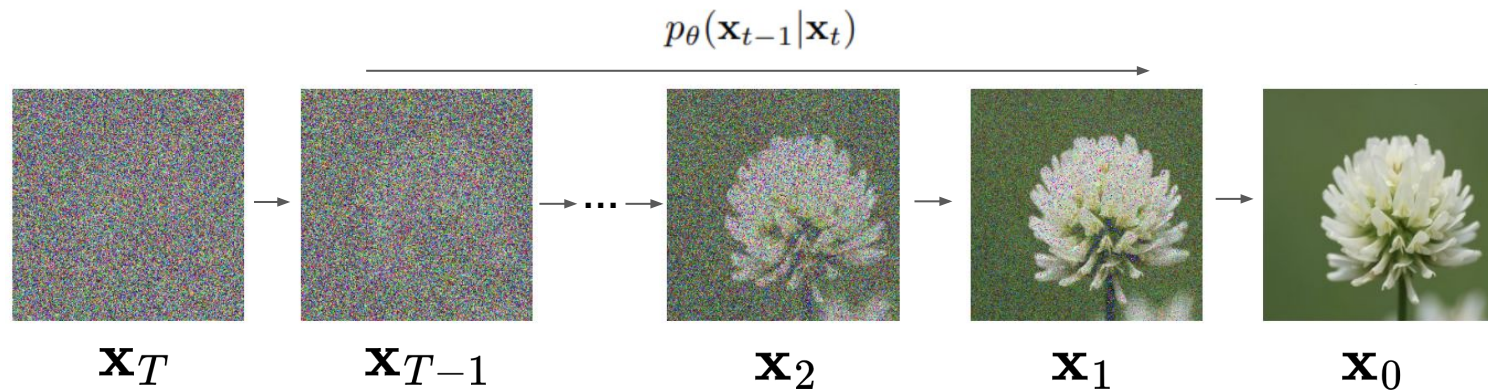


Figure 5. $\bar{\alpha}_t$ throughout diffusion in the linear schedule and our proposed cosine schedule.

Key Idea

We introduce a generative model to approximate the reverse process:

$$\begin{aligned} p(\mathbf{x}_T) &= \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I}) \\ p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I}) \end{aligned} \quad \rightarrow \quad p_{\theta}(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$



Key Idea

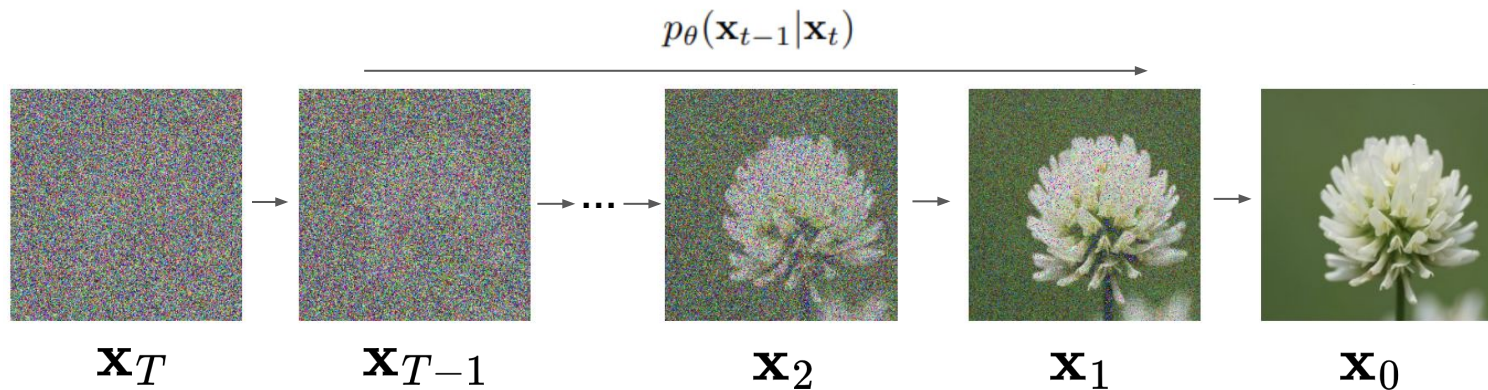
We introduce a generative model to approximate the reverse process:

$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I}) \quad \rightarrow \quad p_{\theta}(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Learning Objective!

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$



Training: Principled Derivation

Find the model that maximizes the likelihood of the training data

i.e. same as VAEs, variational inference; approximate the true posterior

Training Objective

- Bound the likelihood with the ELBO
 - Exactly like VAEs

$$\begin{aligned}\log p(\mathbf{x}) &\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\ &= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}}\end{aligned}$$

Training Objective

- Bound the likelihood with the ELBO
 - Exactly like VAEs

$$\begin{aligned}\log p(\mathbf{x}) &\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\ &= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}}\end{aligned}$$

Discuss: How does the diffusion model construction help minimize the first two terms?

Training Objective

- Bound the likelihood with the ELBO
 - Exactly like VAEs

$$\log p(\mathbf{x}) \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

Learning Objective!

$$= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}}$$

where $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ is the tractable posterior distribution:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

$$\text{where } \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{1 - \beta_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) \text{ and } \tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

Parameterizing the Denoising Model

Since both $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ are Normal distributions, the KL divergence has a simple form:

$$L_{t-1} = D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2 \right] + C$$

Recall that $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{(1 - \bar{\alpha}_t)} \epsilon$. [Ho et al. NeurIPS 2020](#) observe that:

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{1 - \beta_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right)$$

They propose to represent the mean of the denoising model using a **noise-prediction network**:

$$\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1 - \beta_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right)$$

With this parameterization

$$L_{t-1} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\frac{\beta_t^2}{2\sigma_t^2(1 - \beta_t)(1 - \bar{\alpha}_t)} \|\epsilon - \underbrace{\epsilon_\theta(\underbrace{\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}}_{\mathbf{x}_t}, t)}_{\mathbf{x}_t}\|^2 \right] + C$$

Alternative Diffusion Parameterization: Data Prediction

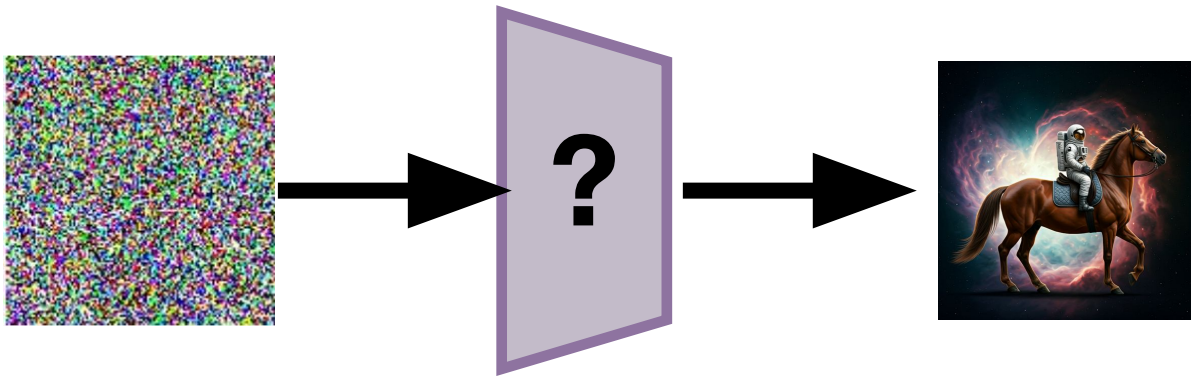
Can also view the diffusion network as learning to predict the **original data**

$$\begin{aligned}\mathbf{x}_t &= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \\ \Rightarrow \mathbf{x}_0 &= \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon}{\sqrt{\bar{\alpha}_t}} \\ \Rightarrow \mathbf{x}_\theta(\mathbf{x}_t, t) &= \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}}\end{aligned}$$

Diffusion Models

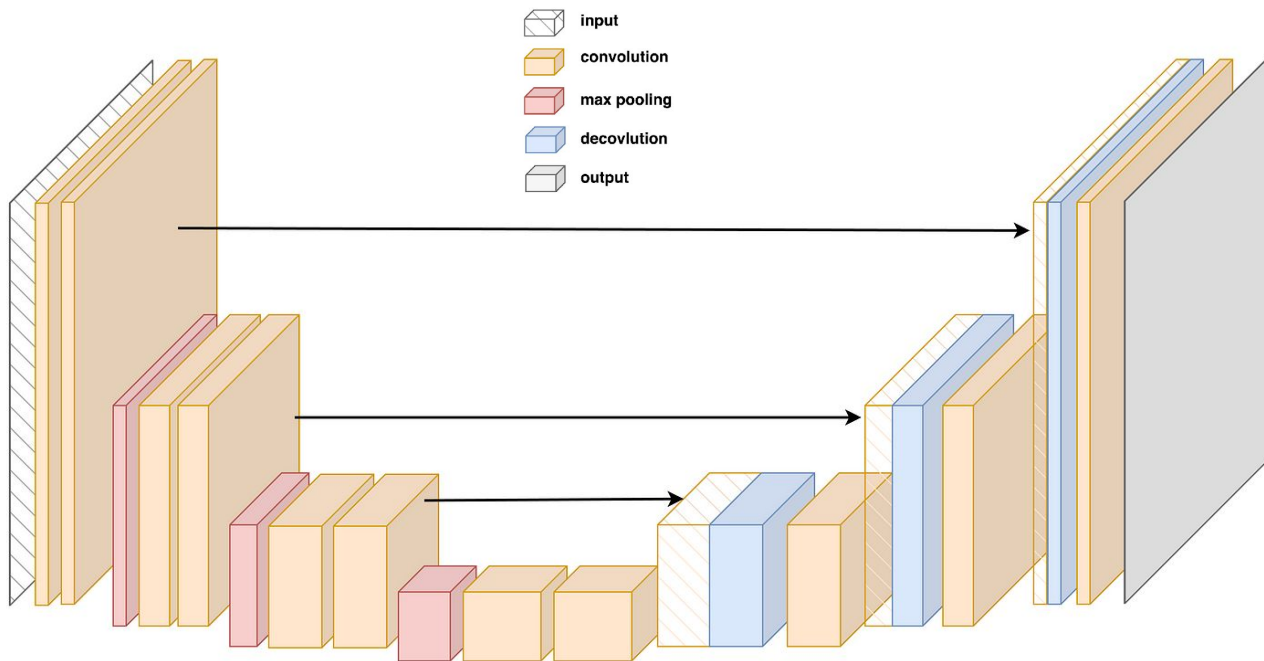
[Sohl-Dickstein et al. 2015]

- Draw a sample of Gaussian noise
- Transform it to obtain a natural image



The U-Net

[Ronneberger et al. 2015]

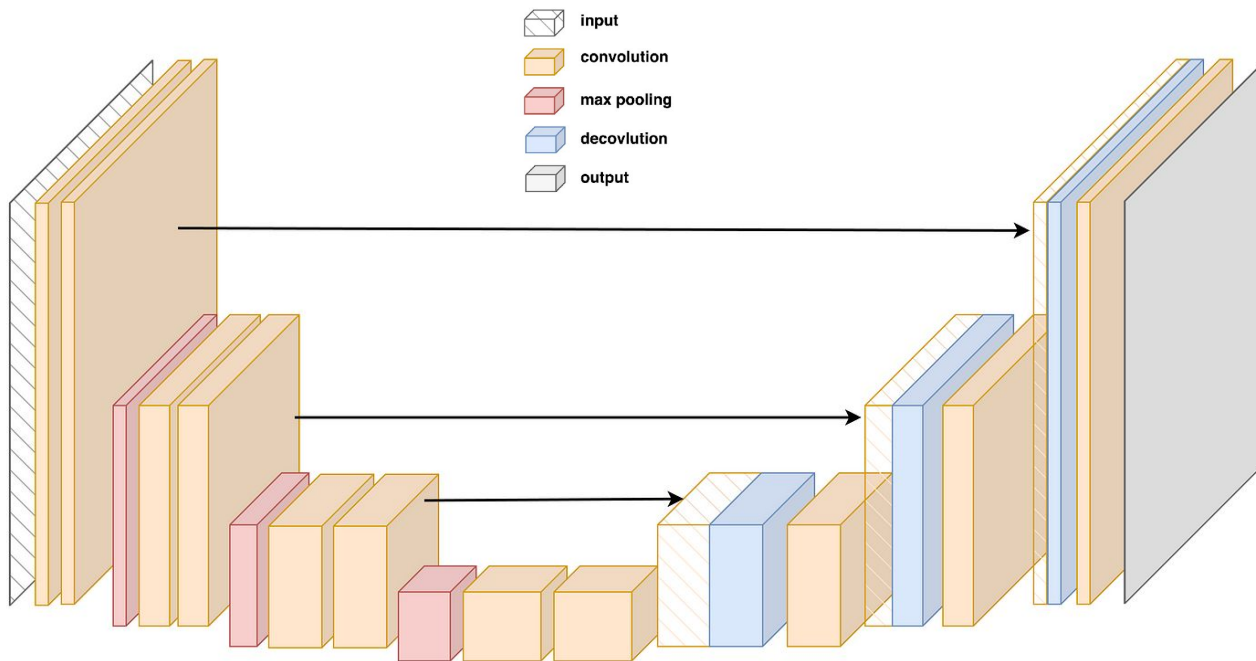


The U-Net

[Ronneberger et al. 2015]

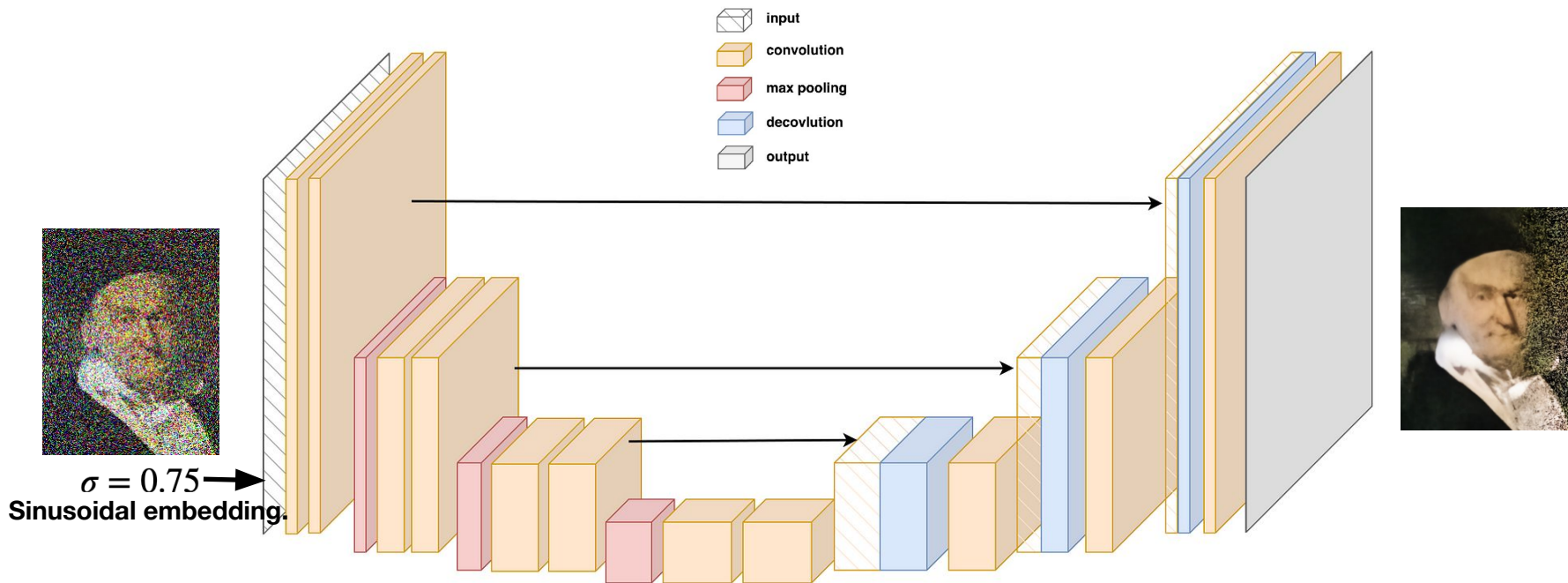


$\sigma = 0.75$



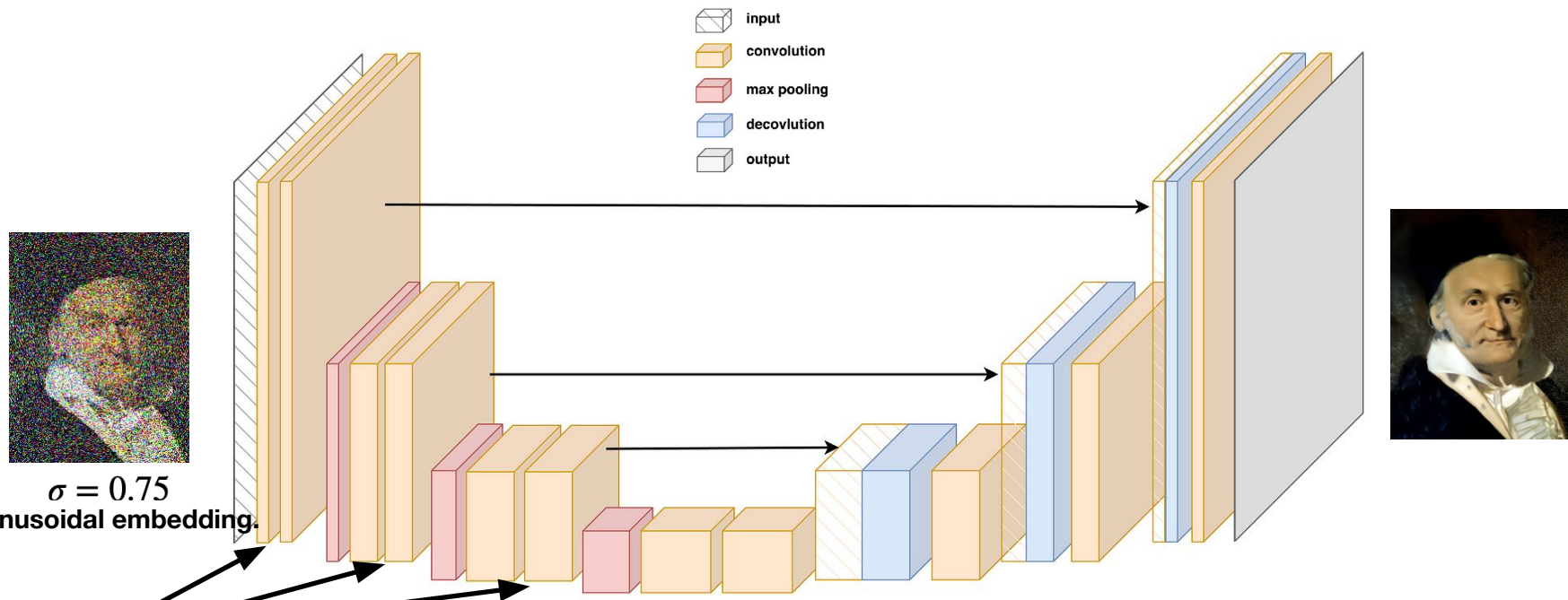
The U-Net

[Ronneberger et al. 2015]



The U-Net

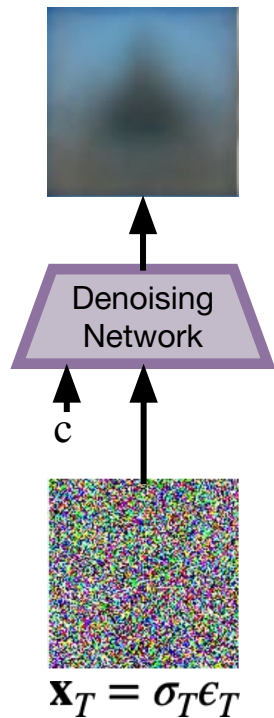
[Ronneberger et al. 2015]



c="An oil painting of a old white man with a white shirt, black hat, and gray side burns."

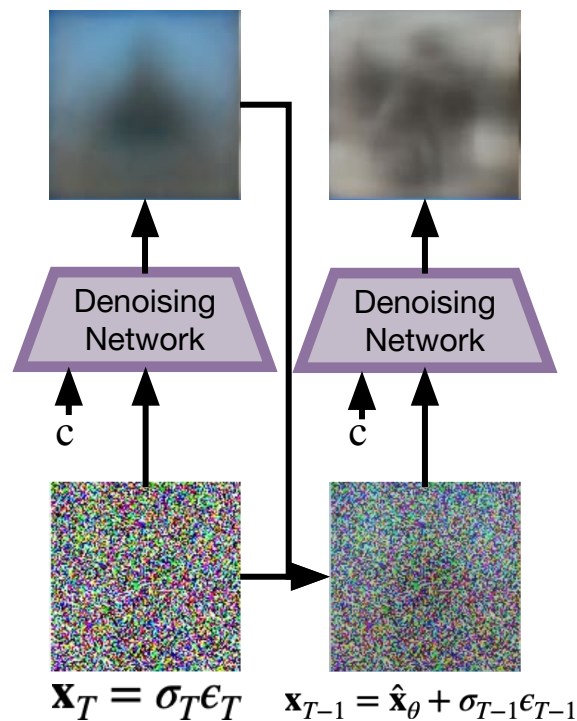
$c = \text{"An astronaut riding a horse"}$

Diffusion Sampling



c="An astronaut riding a horse"

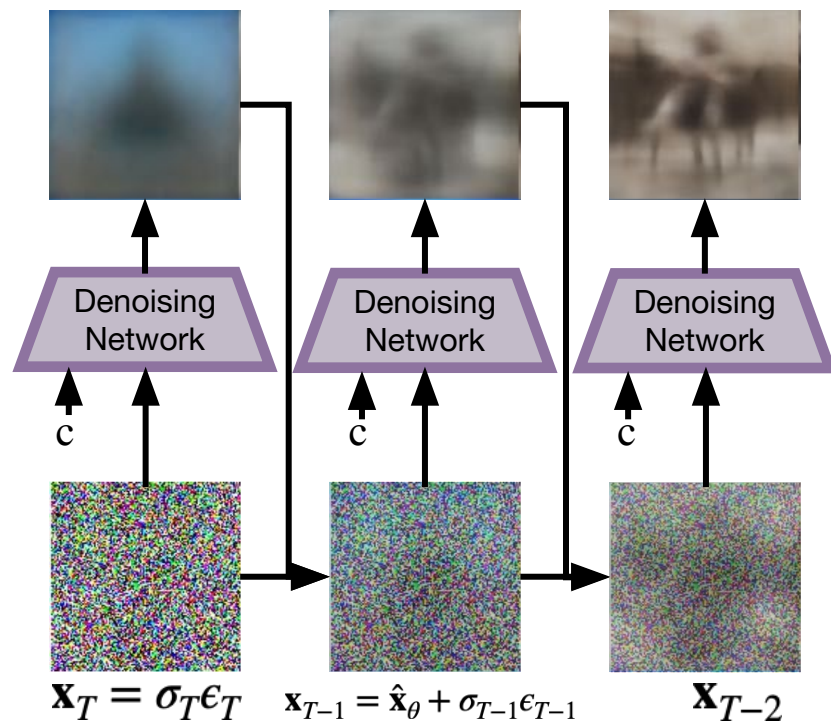
Diffusion Sampling



$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_\theta(\mathbf{x}_t, t)) \approx q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$$

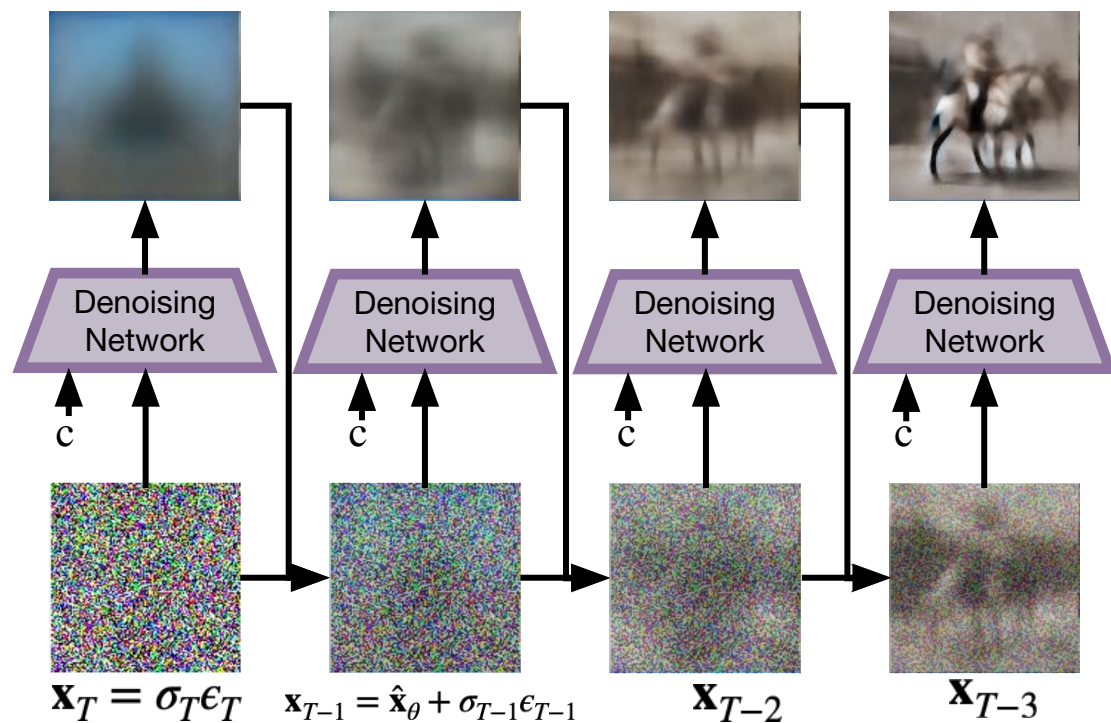
c="An astronaut riding a horse"

Diffusion Sampling



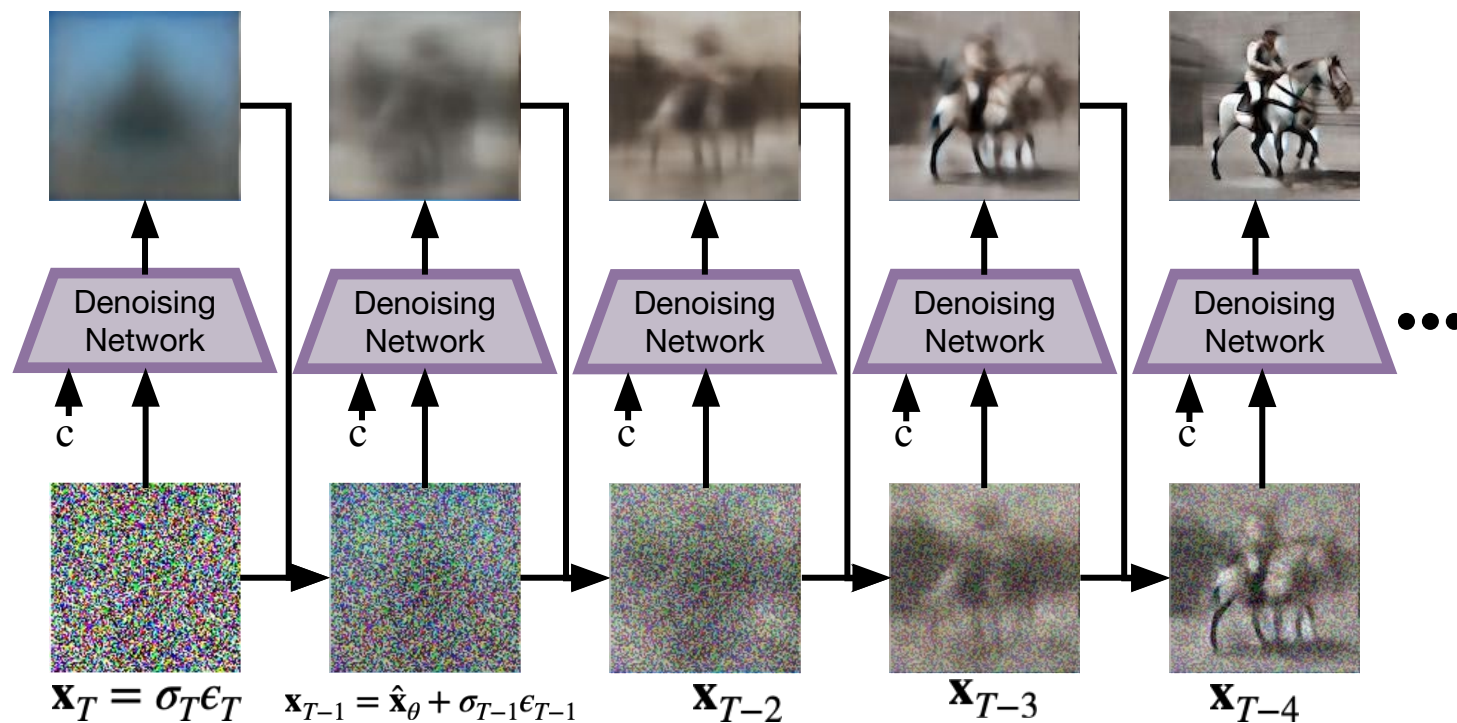
c="An astronaut riding a horse"

Diffusion Sampling



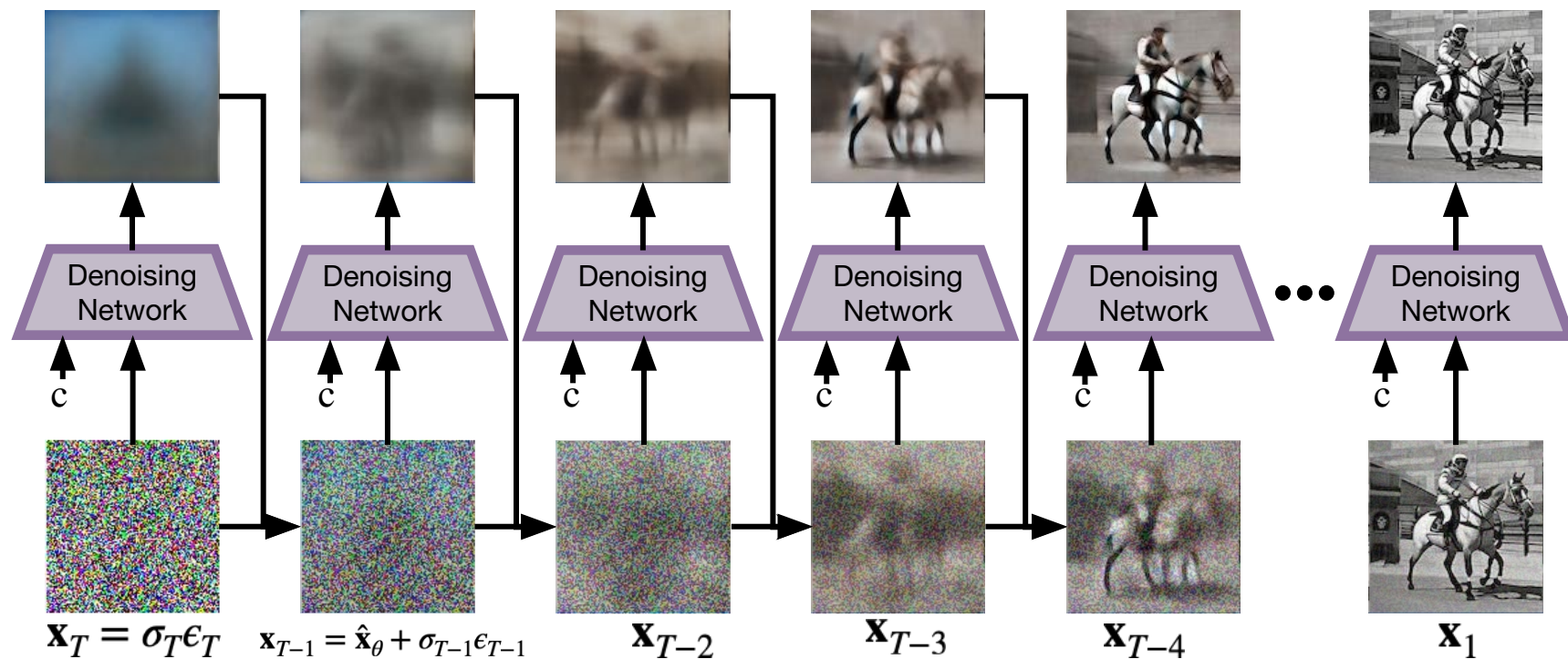
c="An astronaut riding a horse"

Diffusion Sampling



c ="An astronaut riding a horse"

Diffusion Sampling



Diffusion Models



An astronaut riding a horse

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Training Algorithm

Repeat until convergence

1. $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ $\leftarrow$ Sample original image from image distribution
2. $t \sim U\{1, 2, \dots, T\}$ $\leftarrow$ Sample random time step uniformly
3. $\epsilon \sim \mathcal{N}(0, 1)$ $\leftarrow$ Sample Gaussian noise
4. Optimizer step on $L(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2]$
 - $\leftarrow$ Model predicts noise applied at time step t and calculate loss

Inference Sampling Algorithm

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ $\leftarrow$ Sample pure Gaussian noise

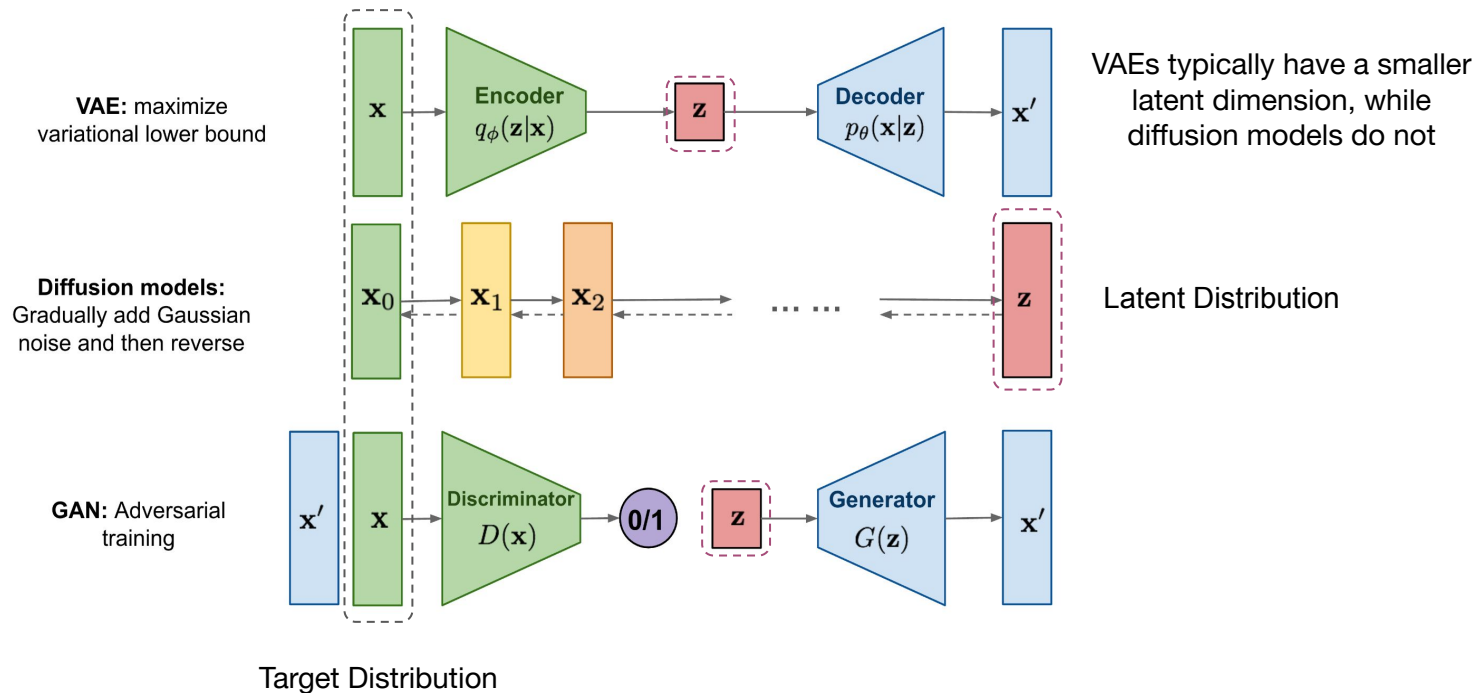
For $t = T, T - 1, \dots, 1$

$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$ $\leftarrow$ Sample Gaussian noise to apply to image

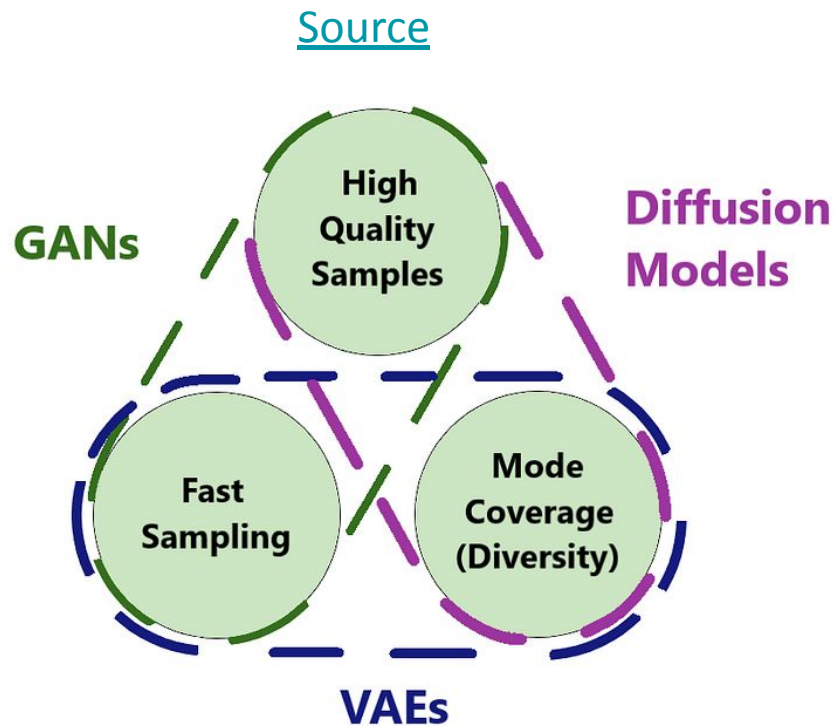
$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ $\leftarrow$ Predict noise applied to image and remove that noise

Return $\mathbf{x}_0$

Generative Modeling



Diffusion Models vs. VAEs vs. GAN



Two Perspectives

Denoising Diffusion Probabilistic Models

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Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decomposition scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at <https://github.com/hojonathanho/diffusion>.

1 Introduction

Deep generative models of all kinds have recently exhibited high quality samples in a wide variety of data modalities. Generative adversarial networks (GANs), autoregressive models, flows, and variational autoencoders (VAEs) have synthesized striking image and audio samples [14, 27, 3, 58, 38, 25, 10, 32, 44, 57, 26, 33, 45], and there have been remarkable advances in energy-based modeling and score matching that have produced images comparable to those of GANs [11, 55].

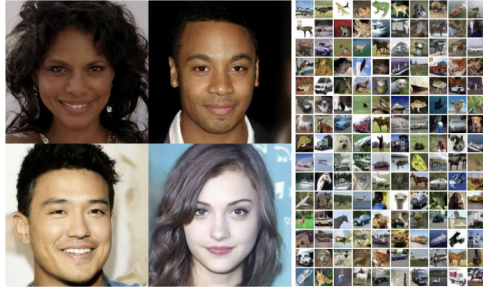


Figure 1: Generated samples on CelebA-HQ 256×256 (left) and unconditional CIFAR10 (right)

Generative Modeling by Estimating Gradients of the Data Distribution

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Abstract

We introduce a new generative model where samples are produced via Langevin dynamics using gradients of the data distribution estimated with score matching. Because gradients can be ill-defined and hard to estimate when the data resides on low-dimensional manifolds, we perturb the data with different levels of Gaussian noise, and jointly estimate the corresponding scores, *i.e.*, the vector fields of gradients of the perturbed data distribution for all noise levels. For sampling, we propose an annealed Langevin dynamics where we use gradients corresponding to gradually decreasing noise levels as the sampling process gets closer to the data manifold. Our framework allows flexible model architectures, requires no sampling during training or the use of adversarial methods, and provides a learning objective that can be used for principled model comparisons. Our models produce samples comparable to GANs on MNIST, CelebA and CIFAR-10 datasets, achieving a new state-of-the-art inception score of 8.87 on CIFAR-10. Additionally, we demonstrate that our models learn effective representations via image inpainting experiments.

1 Introduction

Generative models have many applications in machine learning. To list a few, they have been used to generate high-fidelity images [26, 6], synthesize realistic speech and music fragments [58], improve the performance of semi-supervised learning [28, 10], detect adversarial examples and other anomalous data [54], imitation learning [22], and explore promising states in reinforcement learning [41]. Recent progress is mainly driven by two approaches: likelihood-based methods [17, 29, 11, 60] and generative adversarial networks (GAN [15]). The former uses log-likelihood (or a suitable surrogate) as the training objective, while the latter uses adversarial training to minimize f -divergences [40] or integral probability metrics [2, 55] between model and data distributions.

Although likelihood-based models and GANs have achieved great success, they have some intrinsic limitations. For example, likelihood-based models either have to use specialized architectures to build a normalized probability model (*e.g.*, autoregressive models, flow models), or use surrogate losses (*e.g.*, the evidence lower bound used in variational auto-encoders [29], contrastive divergence in energy-based models [21]) for training. GANs avoid some of the limitations of likelihood-based models, but their training can be unstable due to the adversarial training procedure. In addition, the GAN objective is not suitable for evaluating and comparing different GAN models. While other objectives exist for generative modeling, such as noise contrastive estimation [19] and minimum probability flow [50], these methods typically only work well for low-dimensional data.

In this paper, we explore a new principle for generative modeling based on estimating and sampling from the *(Stein) score* [33] of the logarithmic data density, which is the gradient of the log-density function at the input data point. This is a vector field pointing in the direction where the log data density grows the most. We use a neural network trained with score matching [24] to learn this vector field from data. We then produce samples using Langevin dynamics, which approximately

Score-based Models

Would like to model the probability density function as follows:

$$p_{\theta}(\mathbf{x}) = \frac{e^{-f_{\theta}(\mathbf{x})}}{Z_{\theta}}, \text{ where } Z_{\theta} > 0 \text{ is a normalizing constant s.t. } \int p_{\theta}(\mathbf{x}) d\mathbf{x} = 1$$

Want to maximize the log-likelihood of the data: $\max_{\theta} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}_i).$

Problem: Normalization constant intractable

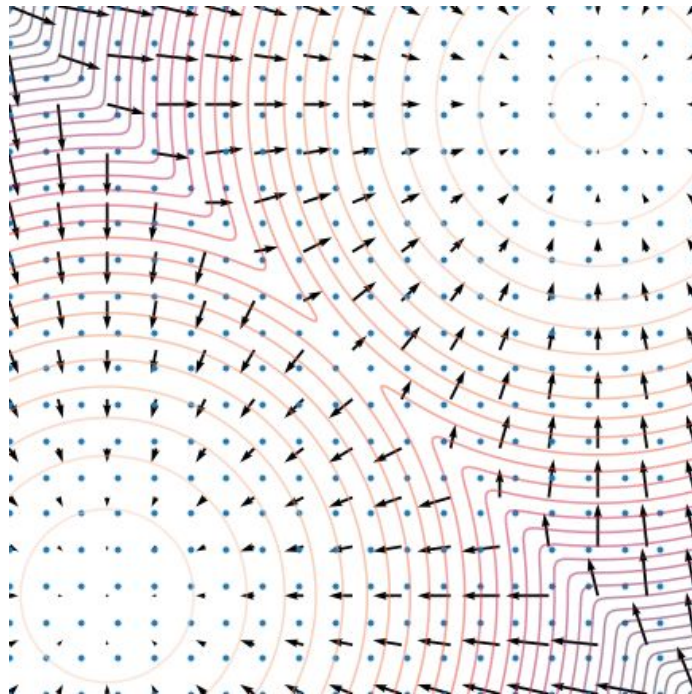
Instead approximate the score function:

$$\mathbf{s}_{\theta}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = -\nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z_{\theta}}_{=0} = -\nabla_{\mathbf{x}} f_{\theta}(\mathbf{x})$$

Score-based Models

Langevin dynamics allow you to sample from distribution - even if it is not normalized.

$$\mathbf{x}_{i+1} \leftarrow \mathbf{x}_i + \epsilon \nabla_{\mathbf{x}} \log p(\mathbf{x}) + \sqrt{2\epsilon} \mathbf{z}_i, \quad i = 0, 1, \dots, K,$$



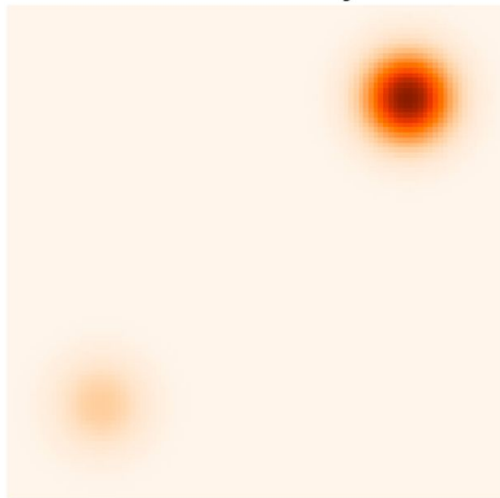
Score Matching

$$\mathbb{E}_{\mathbf{x}} \left[\left\| \boxed{\mathbf{s}_{\theta}(\mathbf{x})} - \boxed{\nabla_{\mathbf{x}} \log p(x)} \right\|_2^2 \right]$$

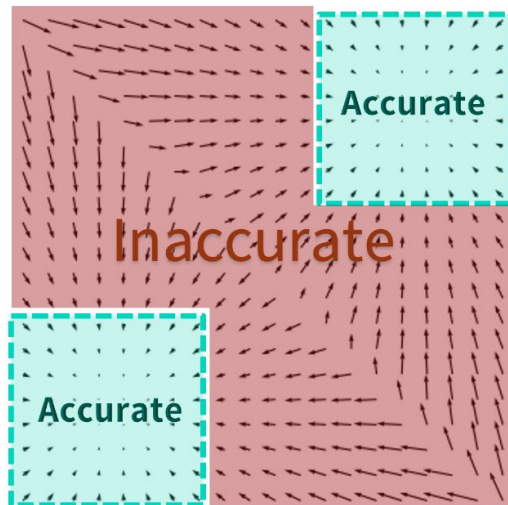
Predicted score!

unknown!

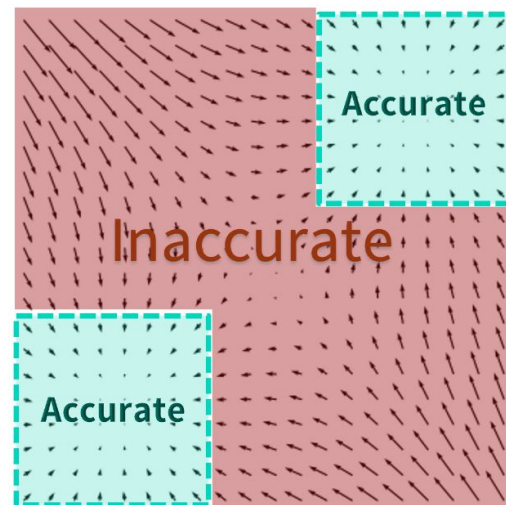
Data density



Data scores

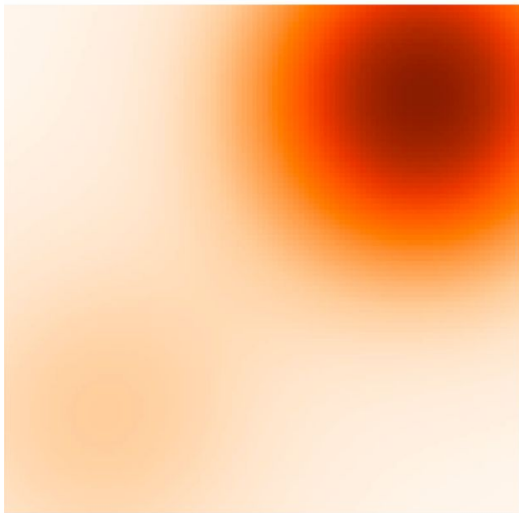


Estimated scores

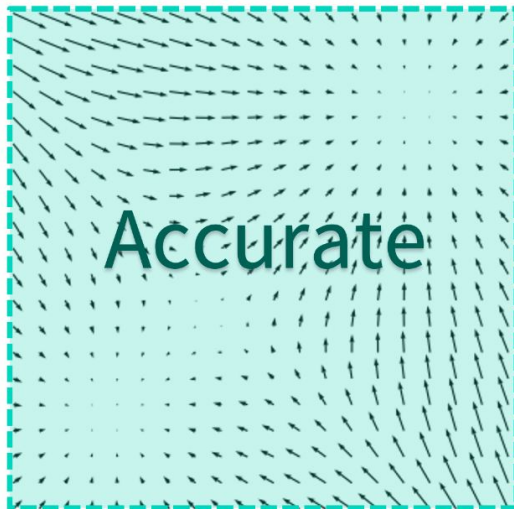


Add noise!

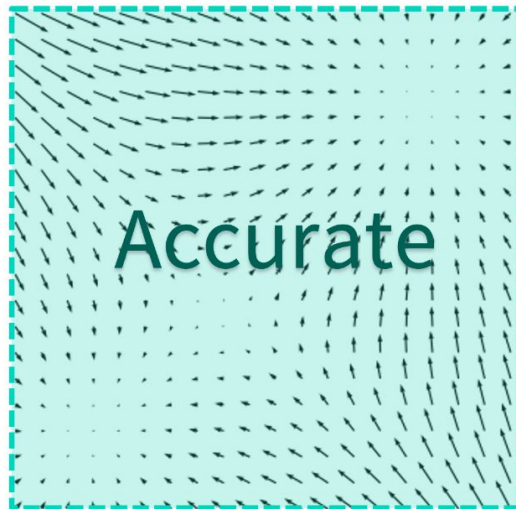
Perturbed density



Perturbed scores



Estimated scores



$$\mathbf{x}_t \sim \mathcal{N}(\mathbf{x}, \sigma_t^2 I)$$

Training

Training Objective for noise level t :

$$\sum_{t=1}^T \lambda(t) \mathbb{E}_{\mathbf{x}, t} [\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)\|_2^2]$$

Using results from denoising score matching [1]:

$$\sum_{t=1}^T \lambda(t) \mathbb{E}_{\mathbf{x}, t} [\|\mathbf{s}_\theta(\tilde{\mathbf{x}}, t) - \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t | \mathbf{x})\|_2^2]$$

Using the definition of the pdf of a Gaussian,

$$\sum_{t=1}^T \lambda(t) \mathbb{E}_{\mathbf{x}, t} [\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}\|_2^2]$$

Looks familiar?

(skip if students look too lost in the sauce ...)

Discuss: How does this training objective relate to that of DDPMs?

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}) &= -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2} \\ &= -\frac{\mathbf{x} + \sigma_t \boldsymbol{\epsilon} - \mathbf{x}}{\sigma_t^2} = -\frac{\sigma_t \boldsymbol{\epsilon}}{\sigma_t^2} = -\boldsymbol{\epsilon} / \sigma_t \\ \implies \mathbf{s}_\theta(\mathbf{x}_t, t) &= -\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) / \sigma_t\end{aligned}$$

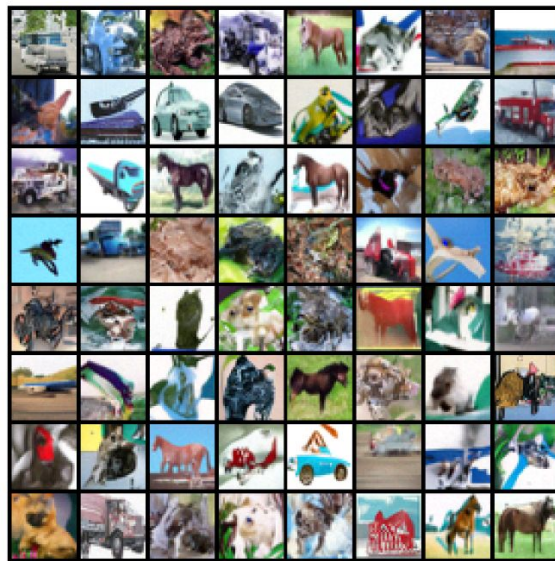
$$\begin{aligned}\lambda(t) \mathbb{E}_{\mathbf{x}, t} [\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}\|_2^2] \\ &= \lambda(t) \mathbb{E}_{\mathbf{x}, t} [\|\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) / \sigma_t - \boldsymbol{\epsilon} / \sigma_t\|_2^2] \\ &= \frac{\lambda(t)}{\sigma_t^2} \mathbb{E}_{\mathbf{x}, t} [\|\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) - \boldsymbol{\epsilon}\|_2^2]\end{aligned}$$



(a) MNIST



(b) CelebA



(c) CIFAR-10

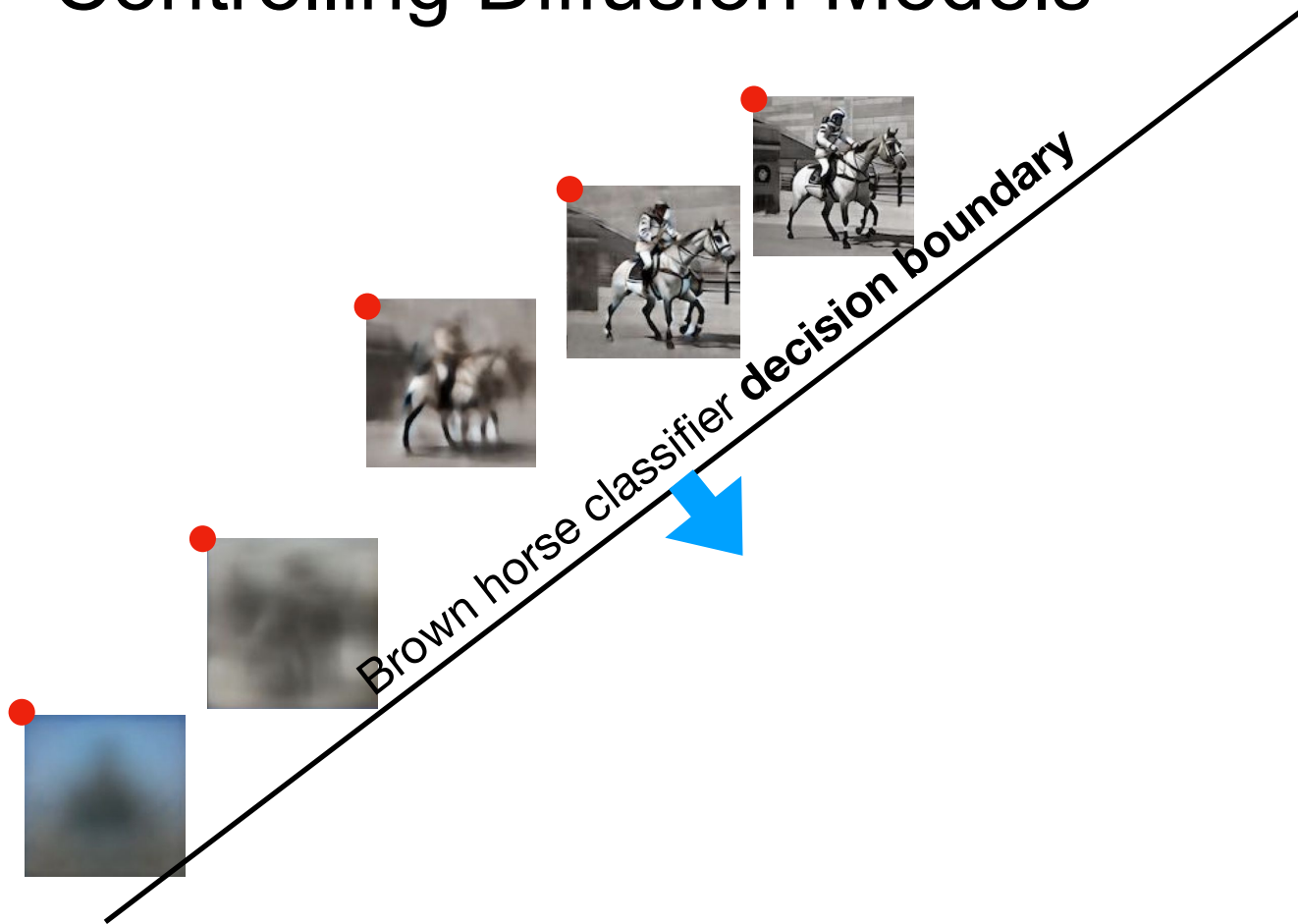
Conditional Diffusion with Classifier Guidance

- May not have access to paired data for training
- Use Bayes' rule to decompose the conditional score into the unconditional score and a likelihood term

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t)$$

- Only need to train a classifier on noised data

Controlling Diffusion Models



Controlling Diffusion Models

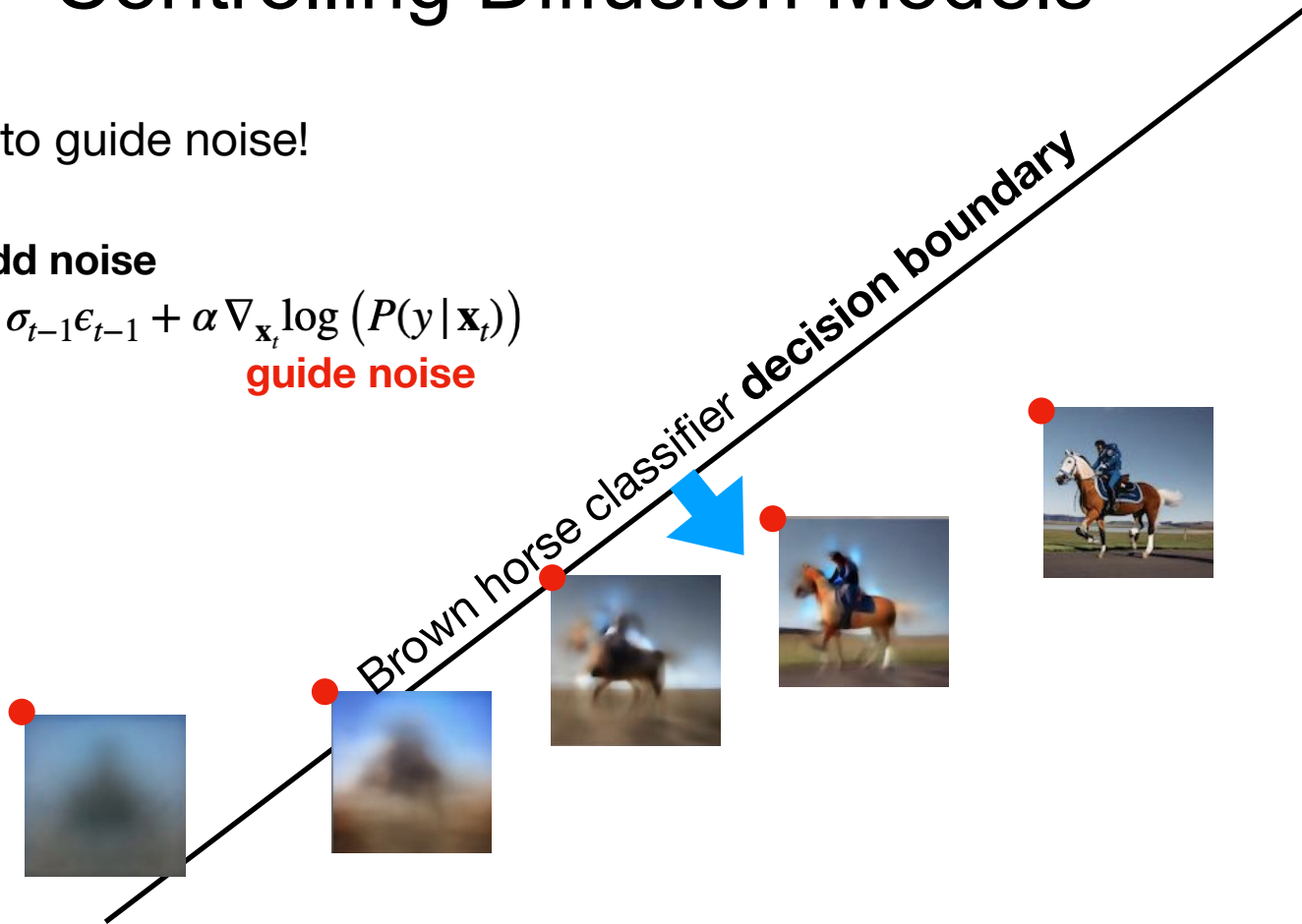
Use classifier to guide noise!

add noise

$$\mathbf{x}_{t-1} = \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t) + \sigma_{t-1}\epsilon_{t-1} + \alpha \nabla_{\mathbf{x}_t} \log(P(y|\mathbf{x}_t))$$

de-noise

guide noise





Unconditional Model



Conditional Model

Classifier-Free Guidance

- Train a joint conditional and unconditional diffusion model
- Conditioning information is added by concatenating to input or cross attending
- Modified conditional distribution

$$\log \tilde{p}_t(\mathbf{x}_t|\mathbf{y}) \propto p_t(\mathbf{x}_t|\mathbf{y}) p_t(\mathbf{y}|\mathbf{x}_t)^w$$

- Conditional sampling

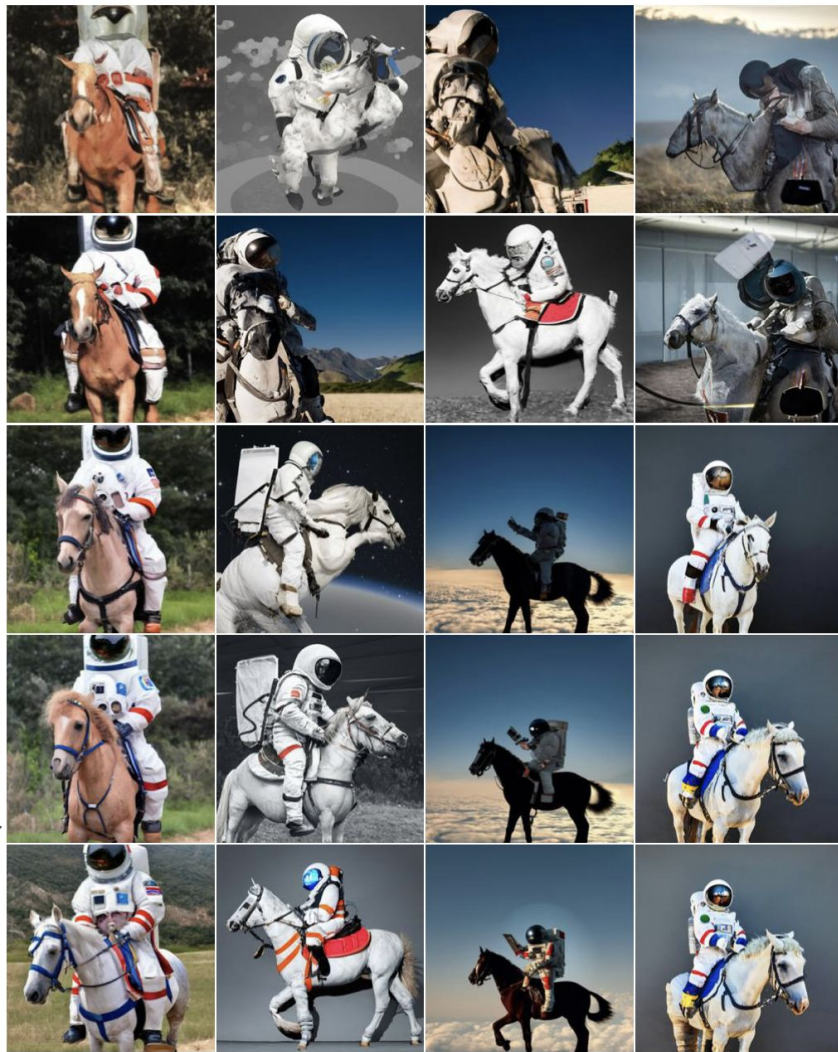
$$\nabla_{\mathbf{x}_t} \log \tilde{p}_t(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + w(\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t))$$

Classifier-Free Guidance

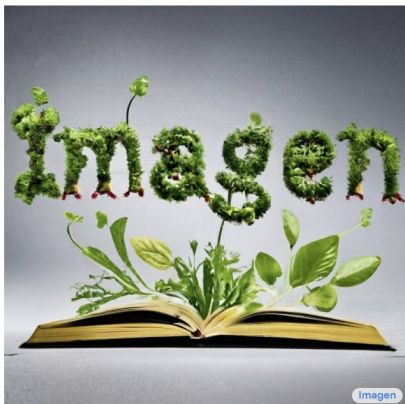
Significantly improves
quality of conditional models

Used by practically **every**
conditional diffusion model

Increasing
Guidance



Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E. L., ... & Norouzi, M. (2022). Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems*, 35, 36479-36494.



Sprouts in the shape of text 'Imagen' coming out of a fairytale book.



A photo of a Shiba Inu dog with a backpack riding a bike. It is wearing sunglasses and a beach hat.



A high contrast portrait of a very happy fuzzy panda dressed as a chef in a high end kitchen making dough. There is a painting of flowers on the wall behind him.



Teddy bears swimming at the Olympics 400m Butterfly event.



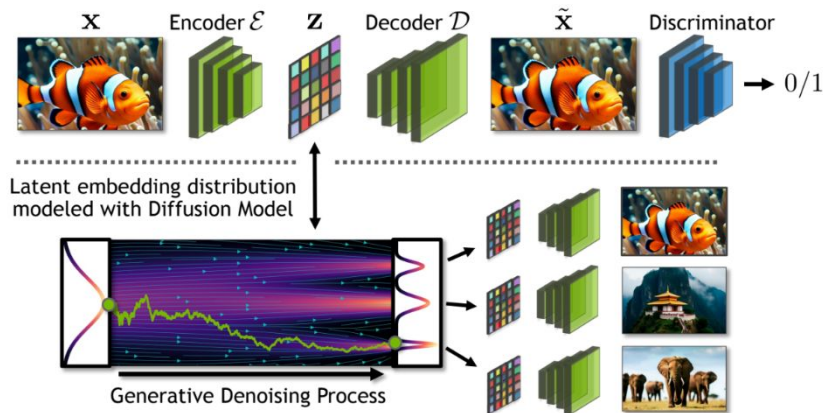
A cute corgi lives in a house made out of sushi.



A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.

Latent Diffusion

Latent Diffusion Models offer Excellent Trade-off between Performance and Compute Demands



- ➔ LDM with appropriate regularization, compression, downsampling ratio and strong autoencoder reconstruction:
- **Computationally efficient** diffusion model in latent space (compression & lower resolution).
 - Yet **very high-performance** (latent diffusion + autoencoder + discriminator = ❤️).
 - **Highly flexible** (can adjust autoencoder for different tasks and data).

Latent Diffusion

Many state-of-the-art large-scale text-to-image models are latent diffusion models

- Stability AI's Stable Diffusion
- Meta's Emu
- OpenAI's Dall-E 3

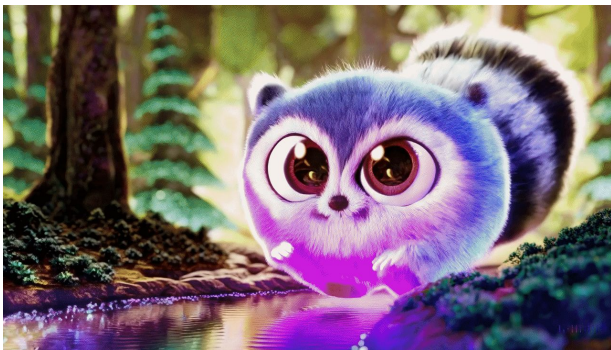
Stable Diffusion Demo!

<https://huggingface.co/spaces/stabilityai/stable-diffusion>

Sample input: "messi as a real madrid player"



Latent Diffusion for Video Generation

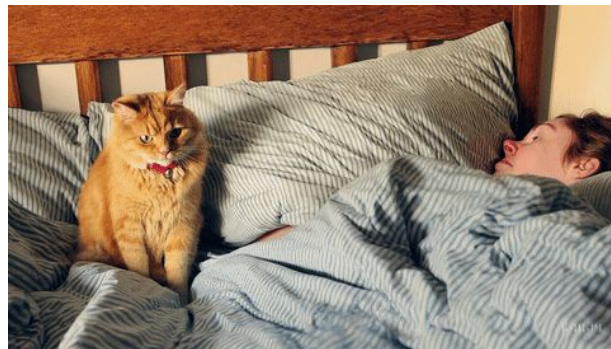


Prompt: 3D animation of a small, round, fluffy creature with big, expressive eyes explores a vibrant, enchanted forest.

Latent Diffusion for Video Generation



Prompt: 3D animation of a small, round, fluffy creature with big, expressive eyes explores a vibrant, enchanted forest.



Prompt: A cat waking up its sleeping owner demanding breakfast.

Review

- Diffusion models can be used to generate high quality samples
- Diffusion was introduced simultaneously from two different perspectives
 - Variational perspective
 - Score-based generative modeling
- **Conditional diffusion** models can be used to generate samples conditioned on other text or image
- **Stable diffusion** is performed in the latent spaces of pre-trained autoencoders

Discrete Diffusion

Large Language Diffusion Models

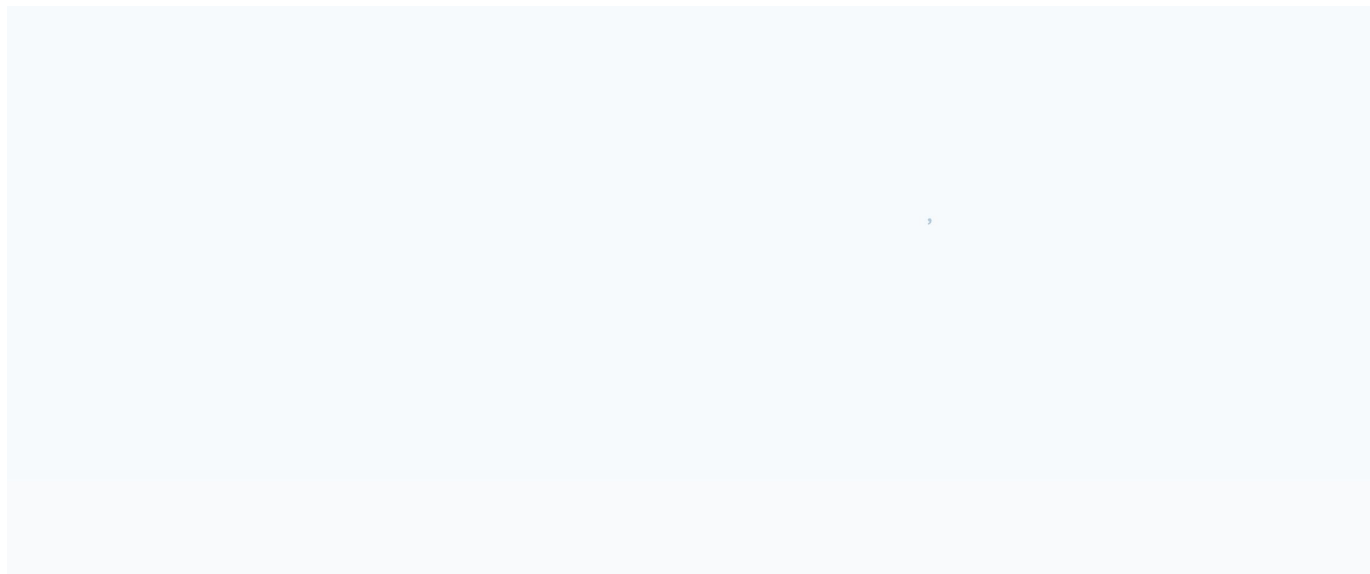
Thanks to: Sofian Zalouk

LLaDA (Large Language Diffusion with mAsking)

Nie et al. (2025)

First paper to scale up discrete diffusion models to 8B params.

Prompt: *Explain what artificial intelligence is.*



Challenging the Autoregressive Framework

Generative Modelling Principle

Key Claim: There is nothing inherently better about left-to-right text generation.

- The key to scalability is maximizing the likelihood of the training data.

$$\max_{\theta} \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log p_{\theta}(x)] \iff \min_{\theta} \text{KL} (p_{\text{data}}(x) \parallel p_{\theta}(x)) .$$

LLaDA demonstrates strong scalability results compared to LLaMA 7B and 8B.

Autoregressive Models

Factorize the joint probability into product of conditional probabilities.

Discrete Diffusion Models

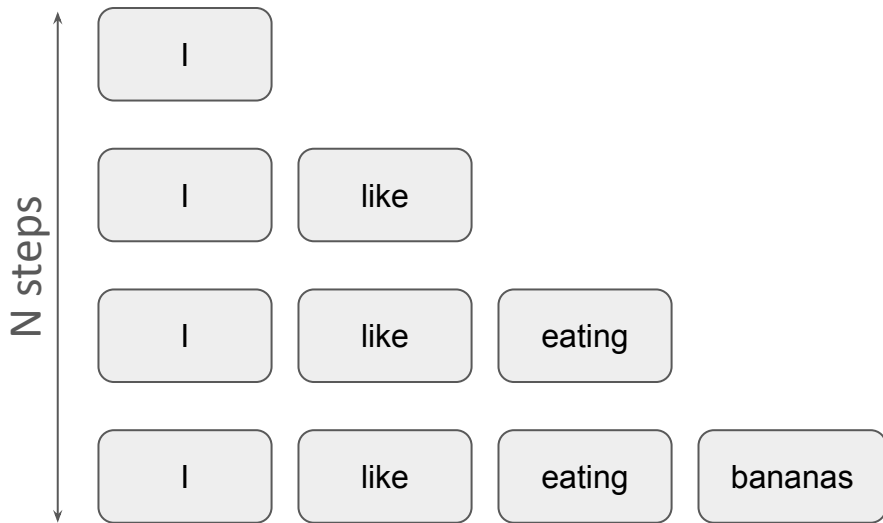
Optimize a lower-bound on likelihood (ELBO)

Challenging the Autoregressive Framework

Beyond Left-to-Right Text Generation

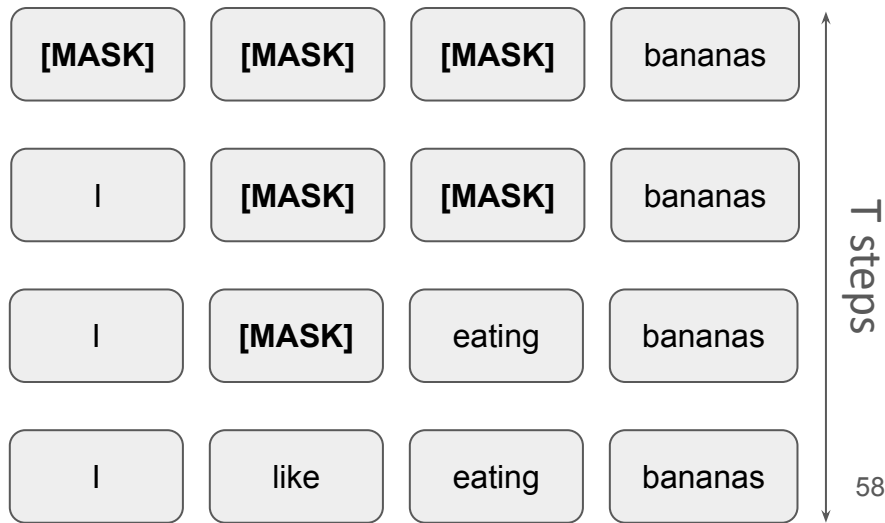
Autoregressive Models

- Next token prediction.
- Left-to-right causal attention.
- One step per token.



Discrete Diffusion Models

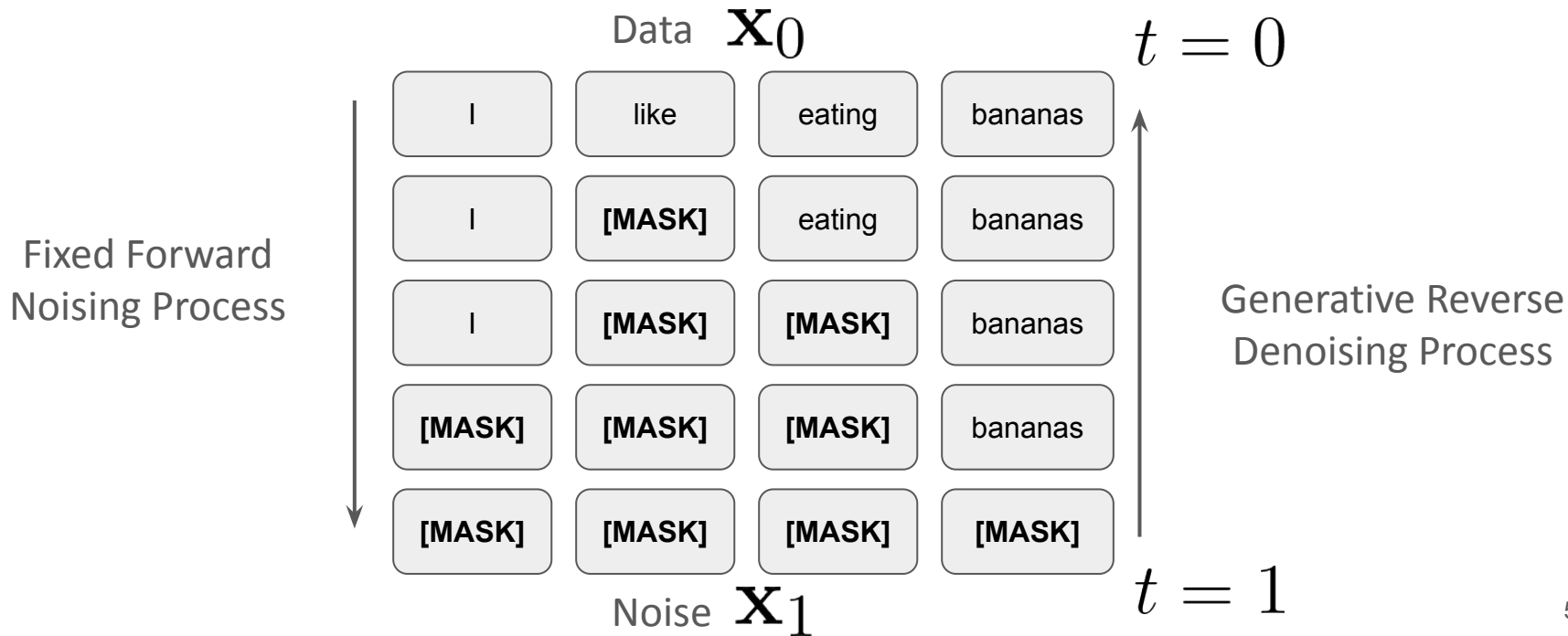
- Any-order token prediction.
- Bi-directional attention.
- T denoising steps per sequence.



Background: Masked Discrete Diffusion

How do we noise text?

Most popular method is by masking tokens independently at random.



Background: Masked Discrete Diffusion

How do we noise text?

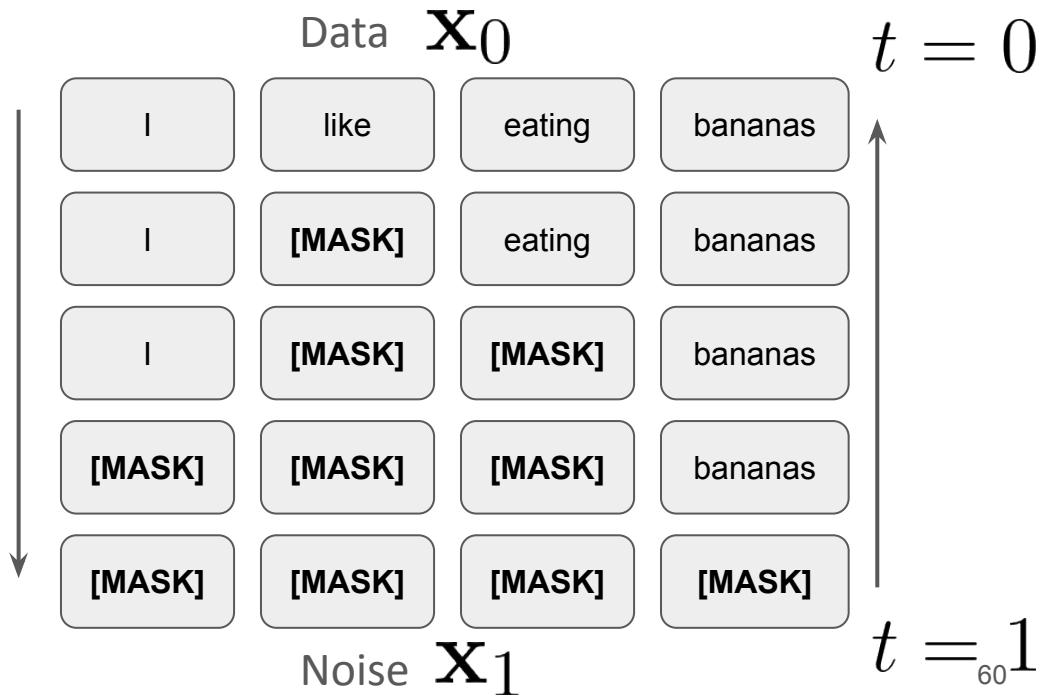
Most popular method is by masking tokens independently at random.

Given a text sequence and $t \in (0,1)$, we can noise it to any intermediate $\mathbf{x}_t$

$$q_{t|0}(x_t^i | x_0^i) = \begin{cases} 1 - t, & x_t^i = x_0^i, \\ t, & x_t^i = \text{M}. \end{cases}$$

Since each token is masked independently:

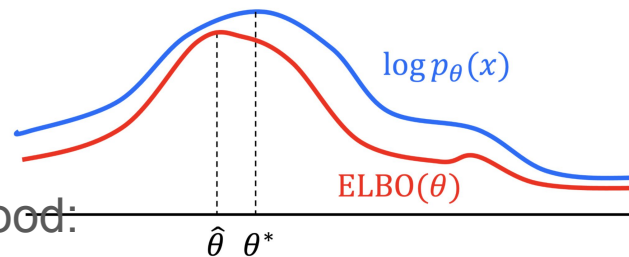
$$q_{t|0}(x_t | x_0) = \prod_{i=1}^L q_{t|0}(x_t^i | x_0^i)$$



Background: Masked Discrete Diffusion

How do we denoise text?

Key idea: Optimize the likelihood of the training data.



In diffusion models, we do not have access to the likelihood:

- Instead, we will optimize the **likelihood lower-bound** (ELBO)

$$\mathbb{E}_{p_{\text{data}}(x_0)}[\log p_{\theta}(x_0)] \leq -\mathbb{E}_{t, x_0, x_t} \left[\frac{1}{t} \sum_{i=1}^L \overset{\text{Masked}}{\mathbf{1}\{x_t^i = \mathbf{M}\}} \log \overset{\text{Model}}{p_{\theta}(x_0^i | x_t)} \right]$$

For Masked Diffusion Models, ELBO has a very intuitive form:

- Given partially masked sequence, model predicts all masked tokens **simultaneously**.
- ELBO computes the likelihood only on the masked tokens.

Background: Masked Discrete Diffusion

How do we generate text?

Key idea: Predict the original text, then add back “noise”.

Given a partially/fully masked sequence $\mathbf{x}_t$, want to generate $\mathbf{x}_s$:

1. Run the discrete diffusion model to simultaneously predict all the masked tokens.
2. For each of the masked tokens, remask them with probability $\frac{s}{t}$

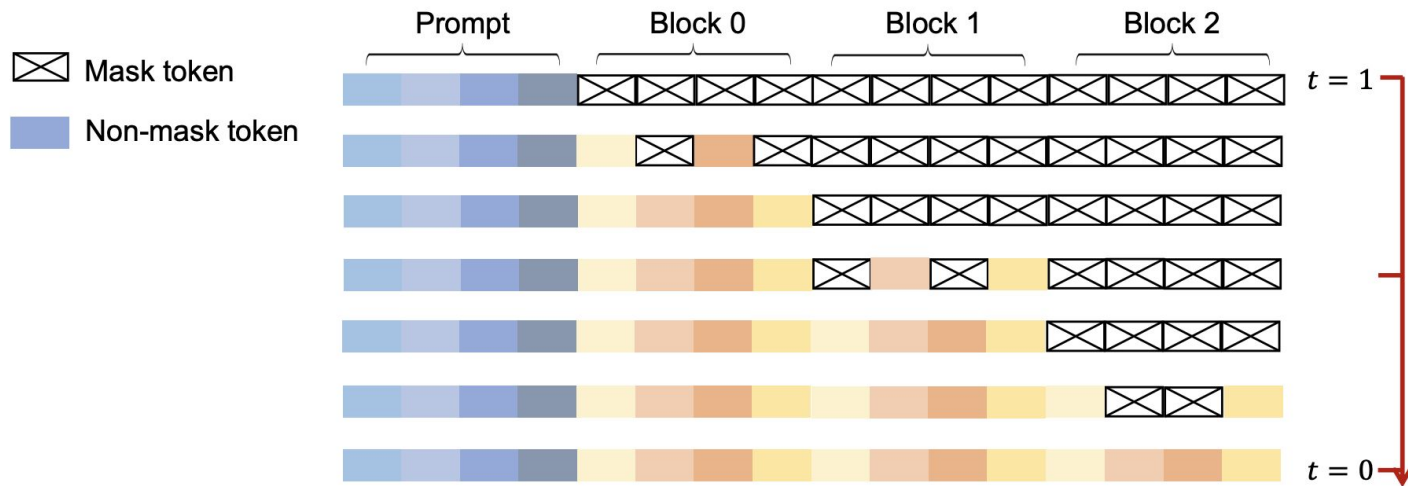
Repeat this process until you generate $\mathbf{x}_0$, the unmasked text.

Implementation Details: Inference

Semi-Autoregressive Sampling

LLaDA applies semi-autoregressive sampling (Arriola et al., ICLR'25)

- Sequence is divided into blocks, generated from left-to-right.
- Within each block, apply LLaDA to unmask/generate.



Implementation Details: Inference

Ablation of Semi-autoregressive sampling and Low-Confidence Remasking

Low-confidence remasking worsens performance of Instruct model:

- SFT training set includes a large number of **|EOS|** tokens.
- Result in high occurrence of **|EOS|** tokens in generation.

Low-confidence + semi-AR fixes this issues → Best accuracy.

Table: GSM8K accuracy

	LLaDA 8B Base	LLaDA 8B Instruct
Randomly remasking	52.3	72.0
Lowest confidence remasking	64.7	12.9
Lowest confidence & semi-autoregressive remasking	64.4	73.8

Results: Multi-Round Dialogue

SFT LLaDA as a Helpful Chat Bot

<i>Multi-round Dialogue Case</i>	
User	Could you please share with me what the initial two lines of the renowned poem ‘The Road Not Taken’?
LLaDA	Certainly! The initial two lines of the renowned poem “The Road Not Taken” by Robert Frost are: “Two roads diverged in a yellow wood, And sorry I could not travel both.” These lines set the stage for the poem, introducing the idea of a speaker facing a choice between two paths.
User	Please help me translate into Chinese.
LLaDA	Sure! The first two lines of “The Road Not Taken” by Robert Frost can be translated into Chinese as: “两条路分岔在黄色的树林中，遗憾我不能同时走”
User	Please translate into German.
LLaDA	Sure! The first two lines of “The Road Not Taken” by Robert Frost can be translated into German as: “Zwei Wege trennten sich im gelben Wald, und leider konnte ich nicht beide tun.”
User	Please also write a poem about life choices. I require 5 sentences, and the first word of each sentence starts with C as the first letter.
LLaDA	Certainly! Here’s a poem about life choices, with each sentence starting with the letter “C”: Choices cascade through life’s journey, Creating connections with each decision, Careful consideration leads to better paths, Courageous choices shape our destiny, Choosing wisely guides us through the years.

Results: Multi-Round Dialogue

Visualization of Generation with Semi-Autoregressive Generation

Darker colors indicate tokens predicted in the later stages of sampling, while lighter colors correspond to earlier predictions

User What are the benefits of regular exercise for physical and mental health?

LLaDA

Regular exercise can improve physical health by strengthening muscles and bones ,
improving cardiovascular health , and reducing the risk of chronic diseases such as
diabetes and heart disease . It can also improve mental health by reducing stress
and anxiety , improving mood , and enhancing cognitive function . Additionally ,
exercise can improve sleep quality and increase energy levels .

Key Challenges

Current Issues and Future Work

Computational Efficiency

- Cannot perform KV caching.
- Cannot use optimizations like grouped-query attention.
- Instead, LLaDA uses multihead attention.
- Decrease dense layer size to compensate.

RLHF Alignment

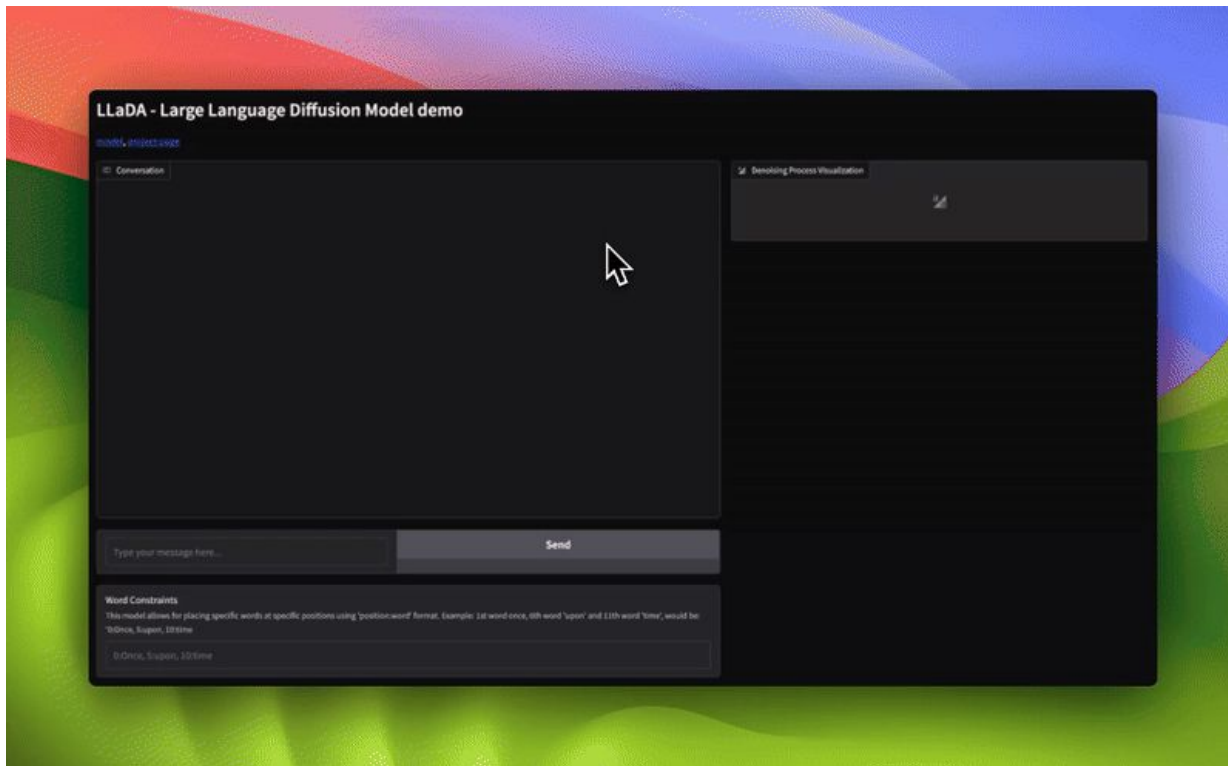
- Unlike autoregressive model, discrete diffusion not compatible with RL alignment.
- Ongoing area of research and an exciting direction for future work.

Once a token is unmasked by LLaDA, it cannot be changed.

- **Solution:** ReMDM (Wang et al. 2025) allows for tokens to be remasked again.
- Discrete Diffusion model can fix its mistakes.

Demo 🍕

<https://huggingface.co/spaces/multimodalart/LLaDA>



<https://chat.inceptionlabs.ai/>